

3.3 Creating and editing charts in MS Excel

During the workshops you will practise creating and editing a variety of different chart types.

“Practise makes perfect” ... or at least, the more you use data visualisation software, the more proficient you will become at making high-quality useful charts, graphs and tables that will help you communicate your message to your audience!

Watch: Excel charts and graphs: Learn the basics for a quick start [14.09 mins]

This video explains how to quickly create graphs for MS Excel reports.

It is useful as it includes suggestions of how to make data comparisons in charts.

As you watch consider:

- How could the suggestions in this video improve the way you create chart visualisations?
- Why is it important to carefully consider how comparisons are visualised?



Source: © Leila Gharani (2017).

Charts, maps and tables

Charts, maps and tables can all be used effectively as data visualisations. It is vital to choose the best one to support the message you want to communicate. The range of charts, maps and tables include the following:

- bar charts
 - clustered bar charts
 - stacked bar charts
- bubble charts
- choropleth maps
- column charts
 - clustered column charts
 - stacked column charts
- funnel charts
- heat maps
- line charts
- scatter charts
- stock charts
- treemaps
- tables or table/chart combinations
- waterfall charts.

□ Watch: Selecting a chart type – part 2 [13.54 mins]

Watch the following video, which introduces various types of maps that can be used to visualise data, outlines when data tables should be used to visualise data and presents some less commonly used chart types.

This is useful as it extends your understanding of the range of available chart types.



[Transcript](#)

(<https://learn.usc.edu.au/courses/35242/files/3138638?wrap=1>)

Source: © University of the Sunshine Coast (2023).

Download the following slides for note taking: [ICT702 Selecting a chart type – part 2](#) (<https://learn.usc.edu.au/courses/35242/files/3172964?wrap=1>)

□ Activity: 2022 World air quality

Access the [2022 World air quality report](#)

(<https://learn.usc.edu.au/courses/35242/files/3138595?wrap=1>) by [IQAir](#) 
(<https://www.iqair.com/ca/>).

As you read take time to consider:

- How easy is it to comprehend the whole report in one reading?
- How might you visualise the content of this report in a short presentation?
- Explain your reasoning and bring your notes to the workshop.

IQAir. (2022). *2022 World air quality report*.

□ Watch: Which is the best chart? [12.07 mins]

This video explains when to use a variety of different chart types and when to avoid using them. This is useful, as knowing what type of chart to avoid is as important as knowing what type of chart may be useful in visualising a given dataset.

Understanding how to convey the most relevant information is vital to conveying the story in the data.

As you watch consider:

- How could the suggestions in this video improve the way you create data visualisations?
- Why is it important to understand when certain chart types are better than others?



Source: © 365 Data Science (2018).

□ Activity: Check your understanding – comparing chart types

In the activity below, chart types descriptions are provided for a range of chart types. Please drag and drop the corresponding chart type to the left of the description which best matches it.

 **Attempt successfully submitted**

Chart type	Description
Bar chart ✓	✓ One of the most frequently used chart types ✓ Composed of a series of bars illustrating how variables develop ✓ Can be confusing if used to compare several variables ✓ Useful to compare one or two variables over time
Pie chart ✓	✓ Circular graph divided into slices ✓ Best suited to depict sections of a whole ✓ Can't show how variables develop over time
Donut chart ✓	✓ Can include a total amount, with a text box in the centre ✓ Can't show how variables develop over time
Line chart ✓	✓ Shows how variable or variables develop over time ✓ Easily highlight the magnitude of change over a period ✓ Can track the development of several variables at the same time ✓ Not useful to show how individual parts change over time ✓ Avoid using stacked line chart, 100% stacked line chart
Area chart ✓	✓ Based on a line chart ✓ Show development of each variable over time ✓ Include regular, stacked and 100% stacked types ✓ Not useful for representing parts of a whole over a single period ✓ Avoid using simple area chart

 **5/5**